U.PORTO
FEUP FACULDADE DE ENGENHARIA UNIVERSIDADE DO PORTO
DEI DEPARTAMENTO DE ENGENHARIA INFORMÁTICA

U.PORTO
FC FACULDADE DE CIÊNCIAS UNIVERSIDADE DO PORTO
[dcc] DEPARTAMENTO DE CIÊNCIA DE COMPUTADORES FACULDADE DE CIÊNCIAS DA UNIVERSIDADE DO PORTO

Lic. in Informatics and Computing Engineering (L.EIC)

Discrete Mathematics

1- Semantics of Propositional Logic - Solutions

1.1) Suppose you are a detective and know the following facts.

- If the knife is in the store room, then we saw it when we cleared the store room.
- The murder was committed at the basement or inside the apartment.
- If the murder was committed at the basement, then the knife is in the yellow dust bin.
- We did not see a knife when we cleared the store room.
- If the murder was committed outside the building, then we are unable to find the knife.
- If the murder was committed inside the apartment, then the knife is in the store room.

The question is: where is the knife?

First, we assigned symbols to the above clues:
s : the knife is in the store room
c : we saw the knife when we clear the store room
b : the murder was committed at the basement
a : murder was committed inside the apartment
y : the knife is in the yellow dust bin
o : the murder was committed outside the building
u : we are unable to find the knife

Formalize the facts as formulas and deduce where the knife is. Create an assignment of truth value for the propositional variables, and confirm that every fact has the truth value **true** for that assignment.

Solution: We want to know where is the knife from these facts. Since they are facts, we assume them to be true. We start by translating the facts into formulas, in the same order as above:

(a) $\mathtt{s} \rightarrow \mathtt{c}$
(b) $\mathtt{b} \vee \mathtt{a}$
(c) $\mathtt{b} \rightarrow \mathtt{y}$
(d) $\neg \mathtt{c}$
(e) $\mathtt{o} \rightarrow \mathtt{u}$
(f) $\mathtt{a} \rightarrow \mathtt{s}$

Now we think of all the assignments that make these facts true. For any such assignment ν:

(a) $\nu(\mathsf{c}) = \mathbf{F}$, because fact 4 is true.

(b) since $\nu(\mathsf{c}) = \mathbf{F}$ then $\nu(\mathsf{s}) = \mathbf{F}$, otherwise fact 1 would not be true.

(c) since $\nu(\mathsf{s}) = \mathbf{F}$ then $\nu(\mathsf{a}) = \mathbf{F}$, otherwise fact 6 would not be true.

(d) since $\nu(\mathsf{a}) = \mathbf{F}$ then $\nu(\mathsf{b}) = \mathbf{T}$, otherwise fact 2 would not be true.

(e) since $\nu(\mathsf{b}) = \mathbf{T}$ then $\nu(\mathsf{y}) = \mathbf{T}$, otherwise fact 3 woulr not be true.

(f) we conclude that "the knife is in the yellow dust bin" since $\nu(\mathsf{y})$ is true in any assignment that makes the facts true.

1.2) Let us consider a propositional language where

- A: "Angelo comes to the party"
- B: "Bruno comes to the party"
- C: "Carlo comes to the party"
- D: "Davide comes to the party"

Formalize the following sentences into a propositional logic formula.

(a) "Angelo comes to the party while Bruno doesn't"

(b) "Either Carlo comes to the party, or Bruno and Davide don't come"

(c) "If Angelo and Bruno come to the party, then Carlo comes provided Davide doesn't come"

(d) "Carlo comes to the party if Bruno and Angelo don't come, or if Davide comes"

(e) "If Angelo comes to the party then Bruno or Carlo come too, but if Angelo doesn't come to the party, then Carlo and Davide come"

Solution:

(a) $\mathsf{A} \wedge \neg \mathsf{B}$

(b) $\mathsf{C} \vee \neg \mathsf{B} \wedge \neg \mathsf{D}$

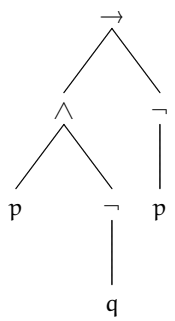
(c) $\mathsf{A} \wedge \mathsf{B} \rightarrow \neg \mathsf{D} \rightarrow \mathsf{C}$

(d) $(\neg \mathsf{A} \wedge \neg \mathsf{B}) \vee \mathsf{D} \rightarrow \mathsf{C}$

(e) $(\mathsf{A} \rightarrow \mathsf{B} \vee \mathsf{C}) \wedge (\neg \mathsf{A} \rightarrow \mathsf{C} \wedge \mathsf{D})$

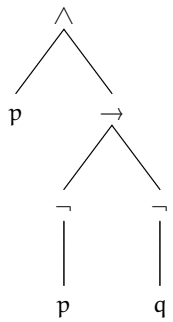
1.3) For each of the following formulas draw its syntactic tree and indicate its subformulas. Remember the priority and associative conventions of logical operations given in the theoretical lecture.
Solutions:

(a) $\mathsf{p} \wedge \neg \mathsf{q} \rightarrow \neg \mathsf{p}$;



Subformulas:
$\mathsf{p} \wedge \neg \mathsf{q}$
$\neg \mathsf{p}$
p
$\neg \mathsf{q}$
q

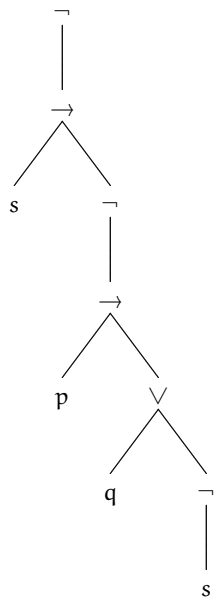
(b) $p \wedge (\neg q \to \neg p)$;



Subformulas:

p
$\neg q \to \neg p$
$\neg q$
$\neg p$
q

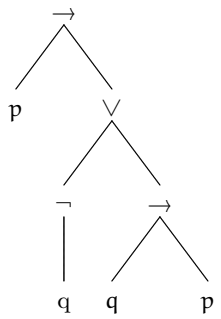
(c) $\neg(s \to (\neg(p \to (q \vee \neg s))))$;



Subformulas:

$s \to (\neg(p \to (q \vee \neg s)))$
s
$\neg(p \to (q \vee \neg s))$
$p \to (q \vee \neg s)$
p
$q \vee \neg s$
q
$\neg s$

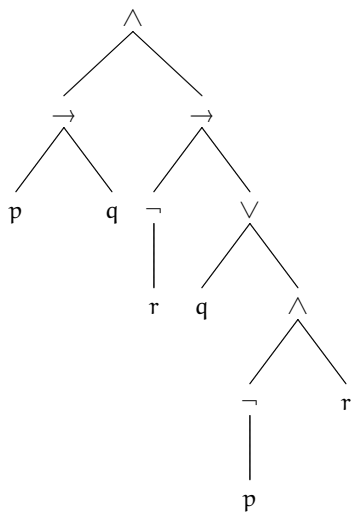
(d) $p \to (\neg q \vee (q \to p))$;



Subformulas:

p
$\neg q \vee (q \to p)$
$\neg q$
q
$q \to p$

(e) $(p \to q) \wedge (\neg r \to (q \vee (\neg p \wedge r)))$.



Subformulas:

$p \to q$
p
q
$\neg r \to (q \vee (\neg p \wedge r))$
$\neg r$
r
$q \vee (\neg p \wedge r)$
$\neg p \wedge r$
$\neg p$

1.4) Construct truth tables for each of the following propositions.
Solutions: We illustrate different ways to present truth tables.

(a) $p \to p$

p	$\to$	p
T	**T**	**T**
T	**T**	**T**
F	**T**	**F**
F	**T**	**F**

The formula is therefore a tautology.

(b) $\neg q \wedge (p \to q)$

p	q	$\neg$	q	$\wedge$	$(p$	$\to$	$q)$
T	**T**	**F**		**F**		**T**	
T	**F**	**T**		**F**		**F**	
F	**T**	**F**		**F**		**T**	
F	**F**	**T**		**T**		**T**	

The formula is therefore satisfiable.

(c) $(p \wedge q) \to \neg(p \vee q)$

p	q	$p \wedge q$	$p \vee q$	$\neg(p \vee q)$	$(p \wedge q) \to \neg(p \vee q)$
T	**T**	**T**	**T**	**F**	**F**
T	**F**	**F**	**T**	**F**	**T**
F	**T**	**F**	**T**	**F**	**T**
F	**F**	**F**	**F**	**T**	**T**

The formula is therefore satisfiable.

(d) $\neg(p \vee q) \to (\neg p \wedge \neg q)$

$\neg$	$(p$	$\vee$	$q)$	$\to$	$(\neg$	p	$\wedge$	$\neg$	$q)$
F	**T**	**T**	**T**	**T**	**F**	**T**	**F**	**F**	**T**
F	**T**	**T**	**F**	**T**	**F**	**T**	**F**	**T**	**F**
F	**F**	**T**	**T**	**T**	**T**	**F**	**F**	**F**	**T**
T	**F**	**F**	**F**	**T**	**T**	**F**	**T**	**T**	**F**

The formula is therefore a tautology.

1.5) When $\nu(p) = \mathbf{T}$ and $\nu(q) = \mathbf{F}$ determine if $\models_\nu \phi$, where ϕ is:

(a) $(p \vee q) \rightarrow (p \wedge q)$;
Solution: From $\nu(p) = \mathbf{T}$ it follows that $\models_\nu p$, therefore $\models_\nu p \vee q$. On the other hand $\nu(q) = \mathbf{F}$ implies that $\not\models_\nu q$, therefore $\not\models_\nu p \wedge q$. Therefore $\not\models_\nu (p \vee q) \rightarrow (p \wedge q)$.

(b) $(p \vee \neg q) \wedge (\neg p \wedge q)$;
Solution: Since $\nu(q) = \mathbf{F}$ then $\not\models_\nu q$. This implies that $\not\models \neg p \wedge q$, from which follows that $\not\models_\nu (p \vee \neg q) \wedge (\neg p \wedge q)$.

(c) $(p \rightarrow q) \wedge (p \rightarrow \neg q)$;
Solution: From $\nu(p) = \mathbf{T}$ and $\nu(q) = \mathbf{F}$ it follows that $\models_\nu p$ and $\not\models_\nu q$, therefore $\not\models_\nu p \rightarrow q$, which implies $\not\models_\nu (p \rightarrow q) \wedge (p \rightarrow \neg q)$.

(d) $p \rightarrow (q \rightarrow (p \vee q))$;
Solution: From $\nu(q) = \mathbf{F}$ it follows that $\not\models_\nu q$, therefore $\models_\nu q \rightarrow (p \vee q))$, which in turn implies $\models_\nu p \rightarrow (q \rightarrow (p \vee q))$.

(e) $((p \rightarrow q) \rightarrow p) \rightarrow p$;
Solution: From $\nu(p) = \mathbf{T}$ it follows that $\models_\nu p$, therefore $\models_\nu ((p \rightarrow q) \rightarrow p) \rightarrow p$.

1.6) Which of the following statements are valid? Justify.

(a) $(p \rightarrow q) \wedge (p \rightarrow r) \Leftrightarrow (p \rightarrow (q \wedge r))$
Solution: There are many ways to show that the statement is valid. We can prove the statement by a sequence of logic equivalences using the equivalence laws of propositional logic.

$$\begin{array}{lcll} (p \rightarrow q) \wedge (p \rightarrow r) & \Leftrightarrow & (\neg p \vee q) \wedge (\neg p \vee r) & (\text{by } \phi \rightarrow \psi \Leftrightarrow \neg\phi \ \vee \ \psi) \\ & \Leftrightarrow & \neg p \vee (q \wedge r) & (\text{using distributivity}) \\ & \Leftrightarrow & (p \rightarrow (q \wedge r)) & (\text{again, by } \phi \rightarrow \psi \Leftrightarrow \neg\phi \ \vee \ \psi) \end{array}$$

Also note that $\varphi \Leftrightarrow \psi$ iff $\varphi \models \psi$ and $\psi \models \varphi$. Therefore the formulas are equivalents if every assignment that satisfies φ also satisfies ψ and vice-versa. This can be verified by checking that the truth tables of both formulas coincide.

$(p$	$\rightarrow$	$q)$	$\wedge$	$(p$	$\rightarrow$	$r)$
T	**T**	**T**	**T**	**T**	**T**	**T**
T	**T**	**T**	**F**	**T**	**F**	**F**
T	**F**	**F**	**F**	**T**	**T**	**T**
T	**F**	**F**	**F**	**T**	**F**	**F**
F	**T**	**T**	**T**	**F**	**T**	**T**
F	**T**	**T**	**T**	**F**	**T**	**F**
F	**T**	**F**	**T**	**F**	**T**	**T**
F	**T**	**F**	**T**	**F**	**T**	**F**

p	$\rightarrow$	$(q$	$\wedge$	$r)$
T	**T**	**T**	**T**	**T**
T	**F**	**T**	**F**	**F**
T	**F**	**F**	**F**	**T**
T	**F**	**F**	**F**	**F**
F	**T**	**T**	**T**	**T**
F	**T**	**T**	**F**	**F**
F	**T**	**F**	**F**	**T**
F	**T**	**F**	**F**	**F**

(b) $(p \rightarrow r) \wedge (q \rightarrow r) \Leftrightarrow ((p \wedge q) \rightarrow r)$
Solution: This statement is not valid. Again this can be verified using truth tables to find an assignment for which the formulas have different truth values. But it is easy to see that, for $\nu(p) = \mathbf{T}$, $\nu(q) = \mathbf{F}$ and $\nu(r) = \mathbf{F}$, then $\not\models_\nu (p \rightarrow r) \wedge (q \rightarrow r)$ because $\not\models_\nu p \rightarrow r$, and $\models_\nu (p \wedge q) \rightarrow r$, since $\not\models_\nu p \wedge q$.

(c) $(p \wedge q) \vee r \Leftrightarrow (p \vee r) \wedge (q \vee r)$
Solution: This statement is the equivalence law for the distributivity of $\vee$ over $\wedge$, therefore it is valid. To prove a basic equivalence law, one uses truth tables.

$(p$	$\wedge$	$q)$	$\vee$	r
T	**T**	**T**	**T**	**T**
T	**T**	**T**	**T**	**F**
T	**F**	**F**	**T**	**T**
T	**F**	**F**	**F**	**F**
F	**F**	**T**	**T**	**T**
F	**F**	**T**	**F**	**F**
F	**F**	**F**	**T**	**T**
F	**F**	**F**	**F**	**F**

$(p$	$\vee$	$r)$	$\wedge$	$(q$	$\vee$	$r)$
T	**T**	**T**	**T**	**T**	**T**	**T**
T	**T**	**F**	**T**	**T**	**T**	**F**
T	**T**	**T**	**T**	**F**	**T**	**T**
T	**T**	**F**	**F**	**F**	**F**	**F**
F	**T**	**T**	**T**	**T**	**T**	**T**
F	**F**	**F**	**F**	**T**	**T**	**F**
F	**T**	**T**	**T**	**F**	**T**	**T**
F	**F**	**F**	**F**	**F**	**F**	**F**

1.7) Without building the truth table, justify the truth or falsity of each of the following statements where p, q, r and s are propositional variables.

(a) $\vDash (p \to (((q \to p) \to p) \to r)$.
Solution: For $\nu(p) = \mathbf{T}$ and $\nu(r) = \mathbf{F}$, we have $\vDash_\nu p$ and $\nvDash_\nu ((q \to p) \to p) \to r$, therefore $\nvDash_\nu (p \to (((q \to p) \to p) \to r))$.

(b) $\vDash (p \to (q \to s)) \to (q \to (p \to s))$.
Solution: This formula is a tautology. Let us first prove this by contradiction. We assume that the formula is not a tautology, that is, there exists an assignment ν that does not satisfy the formula. If $\nvDash_\nu (p \to (q \to s)) \to (q \to (p \to s))$, then $\vDash_\nu p \to (q \to s)$ and $\nvDash_\nu q \to (p \to s)$. From $\nvDash_\nu q \to (p \to s)$ it follows that $\vDash_\nu q$ and $\nvDash_\nu p \to s$, therefore $\vDash_\nu p$ and $\nvDash_\nu s$. But from $\vDash_\nu q$, $\vDash_\nu p$ and $\nvDash_\nu s$ it follows that $\nvDash_\nu p \to (q \to s)$, which leads to a contradiction. This implies that the formula is a tautology.

One can also prove that a formula is a tautology using the laws of equivalence.

$$
\begin{aligned}
&(p \to (q \to s)) \to (q \to (p \to s)) \Leftrightarrow (\neg p \vee (q \to s)) \to (q \to (p \to s)) \\
&\Leftrightarrow (\neg p \vee (\neg q \vee s)) \to (q \to (p \to s)) \Leftrightarrow \neg(\neg p \vee (\neg q \vee s)) \vee (q \to (p \to s)) \\
&\Leftrightarrow \neg(\neg p \vee (\neg q \vee s)) \vee (\neg q \vee (p \to s)) \Leftrightarrow \neg(\neg p \vee (\neg q \vee s)) \vee (\neg q \vee (\neg p \vee s)) \\
&\Leftrightarrow (p \wedge (q \wedge \neg s)) \vee (\neg q \vee (\neg p \vee s)) \Leftrightarrow (p \wedge q \wedge \neg s) \vee \neg q \vee \neg p \vee s \\
&\Leftrightarrow ((p \vee \neg q) \wedge (q \vee \neg q) \wedge (\neg s \vee \neg q)) \vee \neg p \vee s \\
&\Leftrightarrow ((p \vee \neg q \vee \neg p) \wedge (q \vee \neg q \vee \neg p) \wedge (\neg s \vee \neg q \vee \neg p)) \vee s \\
&\Leftrightarrow (p \vee \neg q \vee \neg p \vee s) \wedge (q \vee \neg q \vee \neg p \vee s) \wedge (\neg s \vee \neg q \vee \neg p \vee s)
\end{aligned}
$$

Recall that a formula in conjunctive normal form is a tautology if all disjunctions are tautologies, which is the case of this formula because all disjuntions have a literal and its complement: the first one has p and $\neg p$, the second q and $\neg q$ and the third one s and $\neg s$. Note that if, in addition to the laws of equivalence given in the course we consider the following laws for neutral and absorbing elements, we can effectively show that the formula is equivalent to $\mathbf{T}$.

$$
\begin{array}{lcll}
(\phi \vee \mathbf{F}) & \Leftrightarrow & \phi & \text{(neutral element of } \vee\text{)} \\
(\phi \wedge \mathbf{T}) & \Leftrightarrow & \phi & \text{(neutral element of } \wedge\text{)} \\
(\phi \vee \mathbf{T}) & \Leftrightarrow & \mathbf{T} & \text{(absorbing element of } \vee\text{)} \\
(\phi \wedge \mathbf{F}) & \Leftrightarrow & \mathbf{F} & \text{(absorbing element of } \wedge\text{)}
\end{array}
$$

(c) $\vDash (p \to (q \to r)) \to ((p \to q) \to (p \to r))$.
Solution: Let us proceed by contradiction. We assume that that the formula is not a tautology, therefore there exists an assignment such that $\nvDash_\nu (p \to (q \to r)) \to ((p \to q) \to (p \to r))$, which implies $\vDash_\nu p \to (q \to r)$ and $\nvDash_\nu (p \to q) \to (p \to r)$. But $\nvDash_\nu (p \to q) \to (p \to r)$ implies $\nvDash_\nu p \to r$, therefore $\vDash_\nu p$ and $\nvDash_\nu r$. But then, $\vDash_\nu p \to (q \to r)$ implies that $\nvDash_\nu q$, which then implies that $\nvDash_\nu p \to q$, which leads to $\vDash_\nu (p \to q) \to (p \to r)$, contradicting what was previously stated. Therefore $\vDash (p \to (q \to r)) \to ((p \to q) \to (p \to r))$.

(d) $\vDash ((p \wedge q) \rightarrow (s \vee t)) \rightarrow ((p \rightarrow s) \vee (q \rightarrow t))$.
Solution: In this case let us use semantic equivalence.

$$((p \wedge q) \rightarrow (s \vee t)) \rightarrow ((p \rightarrow s) \vee (q \rightarrow t)) \Leftrightarrow \neg((p \wedge q) \rightarrow (s \vee t)) \vee ((p \rightarrow s) \vee (q \rightarrow t))$$
$$\Leftrightarrow \neg((p \wedge q) \rightarrow (s \vee t)) \vee ((p \rightarrow s) \vee (q \rightarrow t)) \Leftrightarrow \neg(\neg(p \wedge q) \vee (s \vee t)) \vee ((p \rightarrow s) \vee (q \rightarrow t))$$
$$\Leftrightarrow \neg(\neg(p \wedge q) \vee (s \vee t)) \vee ((\neg p \vee s) \vee (\neg q \vee t)) \Leftrightarrow (\neg\neg(p \wedge q) \wedge \neg(s \vee t)) \vee (\neg p \vee s \vee \neg q \vee t)$$
$$\Leftrightarrow (p \wedge q \wedge \neg s \wedge \neg t) \vee \neg p \vee s \vee \neg q \vee t$$
$$\Leftrightarrow ((p \vee \neg p) \wedge (q \vee \neg p) \wedge (\neg s \vee \neg p) \wedge (\neg t \vee \neg p)) \vee s \vee \neg q \vee t$$
$$\Leftrightarrow ((p \vee \neg p \vee s) \wedge (q \vee \neg p \vee s) \wedge (\neg s \vee \neg p \vee s) \wedge (\neg t \vee \neg p \vee s)) \vee \neg q \vee t$$
$$\Leftrightarrow ((p \vee \neg p \vee s \vee \neg q) \wedge (q \vee \neg p \vee s \vee \neg q) \wedge (\neg s \vee \neg p \vee s \vee \neg q) \wedge (\neg t \vee \neg p \vee s \vee \neg q)) \vee t$$
$$\Leftrightarrow (p \vee \neg p \vee s \vee \neg q \vee t) \wedge (q \vee \neg p \vee s \vee \neg q \vee t) \wedge (\neg s \vee \neg p \vee s \vee \neg q \vee t) \wedge (\neg t \vee \neg p \vee s \vee \neg q \vee t)$$

Again, the formula is a tautology since all its disjunctions are tautologies.

1.8) Justify the truth or falsity of each of the following statements, where Γ and Σ represent sets of formulas and $\phi, \psi, \theta, \gamma$ represent formulas of propositional logic. If the statement is false, you must give a counterexample indicating particular values of the sets or formulas for which the statement is false. If the statement is true you must justify using the definitions.

(a) If $\Gamma \vDash \theta$ and $\Gamma \subseteq \Sigma$, then $\Sigma \vDash \theta$.
Solution: The statement is true. To prove, let us assume that $\Gamma \vDash \theta$ and $\Gamma \subseteq \Sigma$ and we want to prove that $\Sigma \vDash \theta$. To prove that $\Sigma \vDash \theta$ we need to show that any assignment ν that satisfies all the formulas in Σ also satisfies θ (if $\vDash_\nu \Sigma$ then $\vDash_\nu \theta$). Let ν be any assignment that satisfies Σ. Since all the formulas in Γ are a part of Σ then ν satisfies Γ, but $\Gamma \vDash \theta$, therefore any assignment that satisfies Γ also satisfies θ, which implies that ν satisfies θ.

(b) If $\Sigma \vDash \theta$ and $\Gamma \subseteq \Sigma$, then $\Gamma \vDash \theta$.
Solution: The statement is false. Let $\Sigma = \{p, p \rightarrow q\}$, $\theta = q$ and $\Gamma = \{p\}$. It is easy to see that $\{p, p \rightarrow q\} \vDash q$ and $\{p\} \subseteq \{p, p \rightarrow q\}$, but $\{p\} \nvDash q$.

(c) If $\Gamma \vDash \theta$ and $\Sigma \vDash \theta$, then $\Gamma \cap \Sigma \vDash \theta$.
Solution: The statement is false. Let $\Sigma = \{p, p \rightarrow q\}$, $\Gamma = \{q\}$ and $\theta = q$. It is easy to see that $\{p, p \rightarrow q\} \vDash q$ and $\{q\} \vDash q$ but $\{p\} \cap \{p, p \rightarrow q\} = \emptyset$ and $\nvDash q$ (q is not a tautology).

(d) If $\Gamma \vDash \theta$ and $\Sigma \subseteq \Gamma$ then $\Sigma \cap \Gamma \vDash \theta$.
Solution: The statement is false. Again let $\Gamma = \{p, p \rightarrow q\}$, $\theta = q$ and $\Sigma = \{p\}$. It is easy to see that $\{p, p \rightarrow q\} \vDash q$ and $\{p\} \subseteq \{p, p \rightarrow q\}$, but $\{p\} \cap \{p, p \rightarrow q\} = \{p\}$ and $\{p\} \nvDash q$.

(e) If $\Sigma \vDash \theta$ then $\Sigma \cap \Gamma \vDash \theta$. **Solution:** The statement is false. Let $\Sigma = \{p\}$, $\theta = p$ and $\Gamma = \emptyset$. Then it is easy to see that $\{p\} \vDash p$ but $\{p\} \cap \emptyset = \emptyset$ and $\nvDash p$ (p is not a tautology).

(f) If $\Sigma \vDash \theta$ and $\Sigma \subseteq \Gamma$ then $\Sigma \cup \Gamma \vDash \theta$.
Solution: The statement is true. Note that, if $\Sigma \subseteq \Gamma$ then $\Sigma \cup \Gamma = \Gamma$, therefore the result follows from (a).

1.9) For each of the truth functions (a), (b) and (c), construct a formula in disjunctive normal form and a formula in conjunctive normal form:

p	q	r	(a)	(b)	(c)
T	**T**	**T**	**T**	**F**	**T**
T	**T**	**F**	**T**	**T**	**F**
T	**F**	**T**	**T**	**F**	**T**
T	**F**	**F**	**T**	**T**	**F**
F	**T**	**T**	**F**	**T**	**T**
F	**T**	**F**	**T**	**T**	**F**
F	**F**	**T**	**T**	**F**	**T**
F	**F**	**F**	**T**	**F**	**F**

Solution: We will used the method studied in the course to define CNFs and DNFs using the truth table.

(a) Disjunctive normal form:

$$(p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r)$$

Conjunctive normal form:

$$(p \vee \neg q \vee \neg r)$$

(b) Disjunctive normal form:

$$(p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r)$$

Conjunctive normal form:

$$(\neg p \vee \neg q \vee \neg r) \wedge (\neg p \vee q \vee \neg r) \wedge (p \vee q \vee \neg r) \wedge (p \vee q \vee r)$$

(c) Disjunctive normal form:

$$(p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge r)$$

Conjunctive normal form:

$$(\neg p \vee \neg q \vee r) \wedge (\neg p \vee q \vee r) \wedge (p \vee \neg q \vee r) \wedge (p \vee q \vee r)$$

1.10) Sets of connectives

(a) Show that the logical connectives $\{\neg, \vee\}$ are a universal (complete) set. That is, propositions of the form $p \to q$, $p \leftrightarrow q$, and $p \wedge q$ can be written using only the connectives $\neg$ and $\vee$.
Solution: First we show that $\to$ is not needed since we have $p \to q \Leftrightarrow \neg p \vee q$. We have also seen in lectures that $p \leftrightarrow q \Leftrightarrow (p \wedge q) \vee (\neg p \wedge \neg q)$. Finally, $(p \wedge q) \Leftrightarrow \neg(\neg p \vee \neg q)$. Which shows that any statement using these connectives can be written using just $\{\neg, \vee\}$.
Note that we have proved in this course that $\{\neg, \wedge, \vee\}$ is a complete set of connectives. So any other complete set of connectives just needs to be able to express any formula involving the connectives in $\{\neg, \wedge, \vee\}$.

(b) Define a new connective $\overline{\wedge}$ as follows:

p	q	$p \overline{\wedge} q$
T	**T**	**F**
T	**F**	**T**
F	**T**	**T**
F	**F**	**T**

Show that the propositions $\neg p$, $p \wedge q$, and $p \vee q$ can be written using only the connective $\overline{\wedge}$. Conclude that $\overline{\wedge}$ is universal (complete).
Solution: We show that $\neg$ is not necessary since $\neg p \Leftrightarrow p \overline{\wedge} p$. We can eliminate $\wedge$, since $p \wedge q \Leftrightarrow \neg(p \overline{\wedge} q) \Leftrightarrow (p \overline{\wedge} q) \overline{\wedge} (p \overline{\wedge} q)$.

p	$p \overline{\wedge} p$
T	**F**
F	**T**

p	q	$\neg$	$(p \overline{\wedge} q)$	$(p \overline{\wedge} q) \overline{\wedge} (p \overline{\wedge} q)$
T	**T**	**T**	**F**	**F**
T	**F**	**F**	**T**	**T**
F	**T**	**F**	**T**	**T**
F	**F**	**F**	**T**	**T**

And $p \vee q \Leftrightarrow \neg(\neg p \wedge \neg q)$, which we have already seen how we can write in terms of $\overline{\wedge}$. So $\{\overline{\wedge}\}$ is complete.

Laws of semantic equivalence

$\phi \wedge \psi$	$\Leftrightarrow$	$\psi \wedge \phi$	(commutativity)
$\phi \vee \psi$	$\Leftrightarrow$	$\psi \vee \phi$	(commutativity)
$\neg(\phi \wedge \psi)$	$\Leftrightarrow$	$(\neg\phi \vee \neg\psi)$	(DeMorgan's Law)
$\neg(\phi \vee \psi)$	$\Leftrightarrow$	$(\neg\phi \wedge \neg\psi)$	(DeMorgan's Law)
$(\phi \wedge \psi) \wedge \theta$	$\Leftrightarrow$	$\phi \wedge (\psi \wedge \theta)$	(associativity)
$(\phi \vee \psi) \vee \theta$	$\Leftrightarrow$	$\phi \vee (\psi \vee \theta)$	(associativity)
$(\phi \vee \phi)$	$\Leftrightarrow$	ϕ	(idempotence)
$(\phi \wedge \phi)$	$\Leftrightarrow$	ϕ	(idempotence)
$(\phi \wedge \psi) \vee \theta$	$\Leftrightarrow$	$(\phi \vee \theta) \wedge (\psi \vee \theta)$	(distributiveness)
$(\phi \vee \psi) \wedge \theta$	$\Leftrightarrow$	$(\phi \wedge \theta) \vee (\psi \wedge \theta)$	(distributivity)
$\neg\neg\phi$	$\Leftrightarrow$	ϕ	(Double negation)
$\phi \rightarrow \psi$	$\Leftrightarrow$	$\neg\phi \vee \psi$	